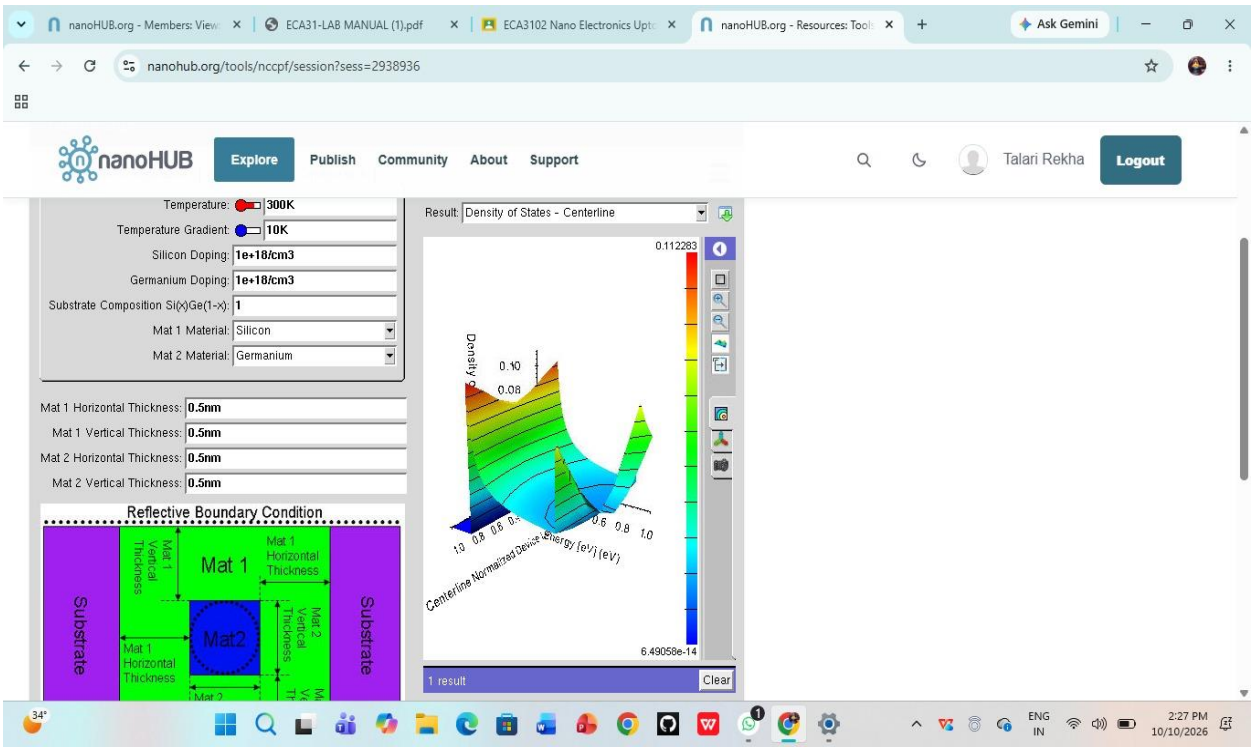
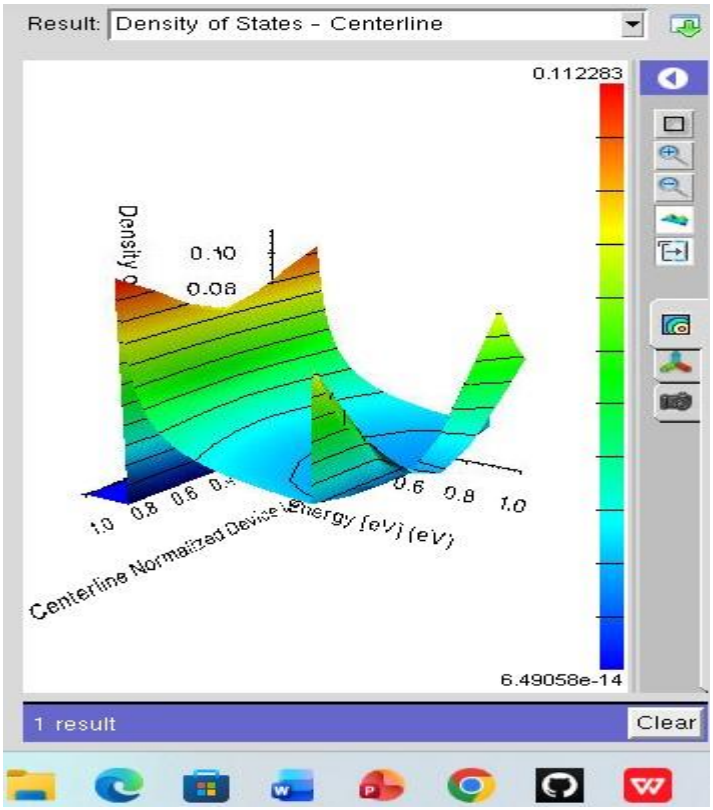
EXPERIMENT 2

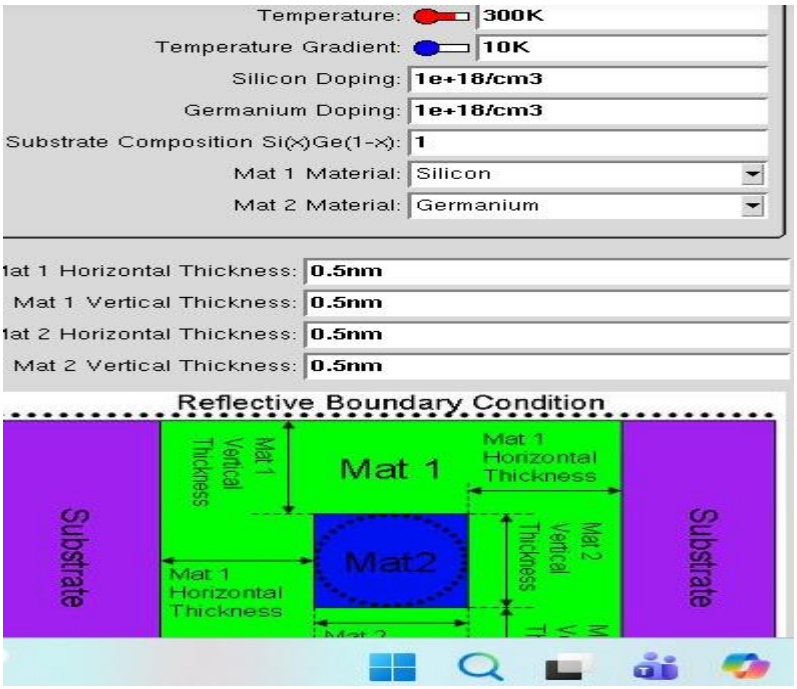
AIM:

Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.

OUTPUT:




Result: Density of States - Centerline
0.112283
Density
0.10
0.08
Centerline Normalized Device Energy [eV] (eV)
6.49058e-14
1 result
Clear


Temperature: 300K
Temperature Gradient: 10K
Silicon Doping: 1e+18/cm3
Germanium Doping: 1e+18/cm3
Substrate Composition Si(x)Ge(1-x): 1
Mat 1 Material: Silicon
Mat 2 Material: Germanium
Mat 1 Horizontal Thickness: 0.5nm
Mat 1 Vertical Thickness: 0.5nm
Mat 2 Horizontal Thickness: 0.5nm
Mat 2 Vertical Thickness: 0.5nm
Reflective Boundary Condition
Substrate
Mat 1 Vertical Thickness
Mat 1
Mat 1 Horizontal Thickness
Mat2
Mat 2 Vertical Thickness
Substrate
Mat 1 Horizontal Thickness